

ADL - Challenge Report

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1 Introduction

This project addresses *pixel-level anomaly detection* in the Spacepresso challenge, where the goal is a continuous *anomaly score map* evaluated with **Pixel-Level Average Precision (AP)**.

Our final solution uses **supervised binary segmentation**:

- one segmentation pipeline per object class;
- real masks plus **online synthetic anomalies**;
- model ensembling with test-time augmentation and post-processing.

2 Problem Analysis

The dataset contains **8 object classes**, including electronic components, mechanical parts and food items. Each sample is represented by **five** 224×224 **views**, and labelled anomalies are provided with pixel-level masks.

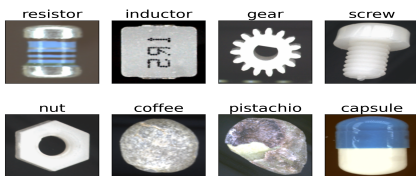


Figure 1: The eight object classes.

The availability of masks makes supervised segmentation possible, but the setting remains strongly data-limited:

1. **Limited supervision.** Each class has about 2,000–2,600 clean images, but only 4–7 labelled anomaly types. Each anomaly type corresponds to a single defective sample.
2. **Defect variability.** Anomalies include scratches, dents, missing parts, stains, contamination and structural deformations, making it hard to rely on a single visual pattern.
3. **Anomalies are sparse.** Only a tiny fraction of pixels are anomalous, so the prediction must stay almost silent elsewhere to avoid drowning the ranking in false positives.

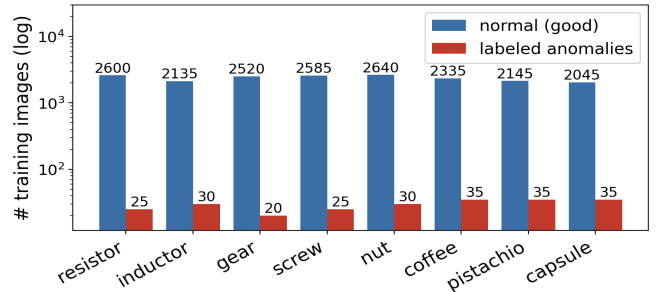


Figure 2: Training images per class (log scale): labelled anomalies are two orders of magnitude rarer than normal images

3 Method

Our final solution treats anomaly detection as **supervised binary segmentation**. For each object class, a model maps an RGB image to a continuous single-channel anomaly score map:

$$\hat{y} = \sigma(f_{\theta}(x)) \in [0, 1]^{H \times W}. \quad (1)$$

This directly matches the pixel-level nature of the evaluation, but requires more positive masks than the dataset provides.

3.1 Online synthetic anomalies

To overcome the lack of labelled defects, we generate additional supervised samples online, following synthetic anomaly approaches such as DRAEM [11], CutPaste [4] and NSA [8]. At each training step, a clean image can be left unchanged, replaced by a real labelled anomaly, or modified with a synthetic defect. Since the modification is created by our pipeline, its binary mask is known exactly.

Table 1: Online data-generation branches and probabilities.

Branch	Prob.
Real anomaly + ground-truth mask	0.20
Good image (empty mask)	0.15
Synthetic line/blob defect	0.25
Synthetic missing chunk	0.20
Texture copy-paste (real anomaly)	0.20

A lightweight object mask restricts synthetic defects to the product area, while smooth noise makes their shape less geometric. Image and mask are then augmented together with flips, small affine transformations, brightness/contrast changes, color jitter and blur.

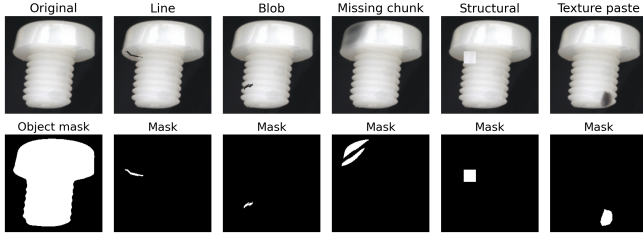


Figure 3: Examples of online synthetic anomaly generation and the corresponding masks used for supervision.

3.2 Architectures and losses

We ensemble three segmentation families to combine different inductive biases: two CNN models and one transformer-based model. The CNNs use a hybrid loss

$$\mathcal{L} = 0.7 \mathcal{L}_{\text{Lovász}} + 0.3 \mathcal{L}_{\text{Focal}}, \quad (2)$$

combining region-level optimisation [2] with hard-pixel focus [5]. SegFormer is trained with BCE+Dice loss. All models use pretrained encoders, AdamW, cosine learning-rate decay, mixed precision and early stopping.

Table 2: Segmentation models used in the final ensemble. The main blend combines the CNN ensemble and SegFormer as $0.55 M_{AB} + 0.45 M_S$.

	A	B	S
Decoder	U-Net [6]	DLv3+ [3]	SegF. [10]
Encoder	EffB4 [9]	EffB5 [9]	MiT-B2
Input	224 ²	224 ²	512 ²
Loss	Hybrid	Hybrid	BCE+Dice
TTA scales	1/1.33/1.66	1/1.33/1.66	1/1.33
Fusion	uncertainty		mean

3.3 Inference

At inference time, flip and multi-scale **test-time augmentation** stabilise predictions. The CNN maps are uncertainty-fused and then blended with SegFormer. We finally smooth the maps, remove tiny components, and apply an HSV foreground mask only outside the product, reducing background false positives while preserving scores on the object. The final maps are encoded in **q8r1e** format.

4 Experiments & Results

We first tested two unsupervised baselines before moving to supervised segmentation. The convolutional autoencoder was trained only on normal images and used the pixel-wise reconstruction error as anomaly score [1]. This reached only ≈ 0.10 Pixel-AP. The result suggests that reconstruction error was not discriminative enough for this dataset: the model often produced smooth reconstructions and the residual maps did not separate defective pixels clearly from normal object variations.

PatchCore [7] was substantially stronger (≈ 0.50 Pixel-AP), showing that pretrained patch features were more effective than raw reconstruction error in our setting. However, its dense score maps were still not reliable enough for the final objective: many normal object regions received high anomaly scores, so the pixel ranking contained too many false positives. This motivated the shift from estimating “distance from normal” to learning defect localisation directly from mask supervision.

Table 3: Pixel-AP across approaches. Baselines are validation estimates; the final score is our best official leaderboard submission.

Approach	Pixel-AP
Convolutional autoencoder	≈ 0.10
PatchCore	≈ 0.50
Supervised ensemble (ours)	0.9264

To motivate the ensemble design, we evaluated each model family independently on 200 synthetic anomalies per class, generated online from normal images with known masks. Table 4 reports the per-class Pixel-AP. The three families achieve similar aggregate scores (0.96 ± 0.01), but their activation maps are not identical, suggesting that they make complementary errors. Combining them therefore improves robustness across the diverse defect types present in the dataset.

Table 4: Per-class Pixel-AP of individual models on synthetic anomalies. Similar aggregate scores but complementary qualitative behaviour motivate the ensemble.

Class	A	B	S
class_01	0.9494	0.9436	0.9518
class_02	0.9664	0.9537	0.9459
class_03	0.9498	0.9444	0.9504
class_04	0.9696	0.9653	0.9605
class_05	0.9622	0.9571	0.9617
class_06	0.9794	0.9803	0.9699
class_07	0.9492	0.9625	0.9331
class_08	0.9533	0.9552	0.9593
Mean	0.9599	0.9578	0.9541

These results motivated the final segmentation

strategy: instead of only modelling normality, the model learns defect localisation directly from real and synthetic masks. Figure 4 shows a representative real defect. The individual model families already localise the anomalous region, while the fused prediction gives a cleaner and sharper score map.

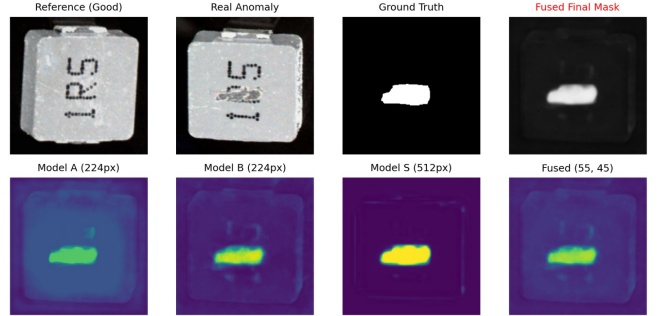


Figure 4: Qualitative result on a real anomaly.

5 Discussion

Strengths. Optimising the pixel objective directly, fed by unlimited mask-paired synthetic supervision, unlocked the large jump in AP; the diverse ensemble and sparsity-aware post-processing add robustness across very different defect types.

Limitations. (i) Validation minimised training *loss* rather than Pixel-AP directly, so model selection relied on an imperfect proxy; a held-out AP-based split would give more faithful signal. (ii) Models A and B operate at 224^2 , which may miss fine-grained or thin defects better captured at higher resolution. (iii) The HSV foreground mask is a heuristic that can suppress true defects near the object boundary under non-standard lighting.

6 Conclusions

We turned an unsupervised-by-default problem into a tractable **supervised segmentation** task by **generating anomalies and their masks online**, reaching an official leaderboard Pixel-AP of **0.9264**. Promising directions include training models A and B at 512^2 to match SegFormer’s input resolution, adopting a held-out anomaly split for AP-based model selection, and exploring per-class post-processing calibration.

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